ACHINTYA NARULA

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MACHINE LEARNING | DATA SCIENCE | PYTHON | SQL | NLP

PROFILE

Final-year B.Tech CSE student with an AI & ML minor and project experience in Python machine learning, NLP, and SQL data pipelines. Built workflows for preprocessing, cross-validation, explainability, FastAPI inference, and dimensional ETL. Seeking applied AI/ML, Python, and data internships from January 2027.

EDUCATION

Shaheed Bhagat Singh State University **Expected May 2027**

B.Tech, Computer Science and Engineering | Minor: Artificial Intelligence & Machine Learning

CGPA: 7.53/10 | Relevant Coursework: Artificial Intelligence, Machine Learning, Data Mining & Data Warehousing, DBMS, Algorithms

AI / ML & DATA SKILLS

Machine Learning: scikit-learn, XGBoost, Logistic Regression, Random Forest, classification, preprocessing pipelines, feature engineering, 5-fold stratified cross-validation, GridSearchCV

Evaluation & Explainability: ROC-AUC, PR-AUC, precision, recall, F1, confusion matrices, out-of-fold threshold selection, SHAP, permutation importance, leakage prevention

Python & Data Analysis: Python, pandas, NumPy, SQL, T-SQL, Joblib; data cleaning, exploratory data analysis (EDA), preprocessing, structured-data workflows, model serialization

Data Engineering & Analytics: SQL Server, ETL, staging, star schema, fact/dimension modeling, surrogate keys, Date Dimension, SCD Type 1 MERGE, CTEs, window functions

NLP & Deployment: TF-IDF, word/bigram features, text classification, FastAPI, Docker, pytest, REST APIs, JSON, Git, GitHub, GitHub Actions

SELECTED MACHINE LEARNING & DATA PROJECTS

Customer Churn Prediction & Explainability | Python, scikit-learn, XGBoost, SHAP

• Built a preprocessing and model-selection pipeline on IBM's 7,043-row Telco sample; compared Logistic Regression, Random Forest, and XGBoost with training-only 5-fold stratified GridSearchCV.

• Selected XGBoost using cross-validated ROC-AUC; recorded 0.8486 ROC-AUC and 0.6585 PR-AUC on an untouched 1,761-row holdout, with precision, recall, F1, and confusion-matrix analysis.

• Added training out-of-fold threshold selection, SHAP, permutation importance, Joblib persistence, FastAPI inference, Docker serving, automated tests, and GitHub Actions CI.

Sales Analytics & Data Warehouse Pipeline | SQL Server, T-SQL, Python, pandas

• Built a SQL Server star schema and SCD Type 1 MERGE loads using deterministic fictional data: 500 customers, 60 products, 5,000 sales, and a 30-customer update batch.

• Added data-quality checks, analytical SQL, and five DAX measures; SQL Server integration CI verifies 30 initial updates, zero repeated updates, unchanged customer counts, no duplicates, and 5,000 fact rows.

IssueSense | NLP Text Classification | Python, pandas, scikit-learn, TF-IDF

• Built TF-IDF word/bigram features and Logistic Regression for bug, feature, and documentation labels; achieved 82.6% accuracy and 82.7% macro-F1 on a 23-row holdout from 90 curated demo examples.

• Added confidence-based review routing, per-class metrics, Joblib persistence, a CLI, JSON reports, and data validation; 15 automated tests run in GitHub Actions.

ADDITIONAL EXPERIENCE

Full-Stack Development Intern | Solitaire Infosys **Jul 2025 - Aug 2025**

• Built application features and API/data integrations with React, Node.js, Express, and MongoDB in a collaborative development environment; received the Best Intern Award.

Web Development Intern | ShadowFox **May 2025 - Jun 2025**

• Completed REST API integration, testing, and Git-based development tasks, strengthening practical programming and data-flow fundamentals.

CERTIFICATIONS & LEADERSHIP

Python for Everybody Specialization - University of Michigan | Crash Course on Python - Google | Power BI Fundamentals

Technical Head, GDG On Campus, SBSSU - organize developer workshops, hackathons, hands-on GenAI labs, and peer-learning activities.